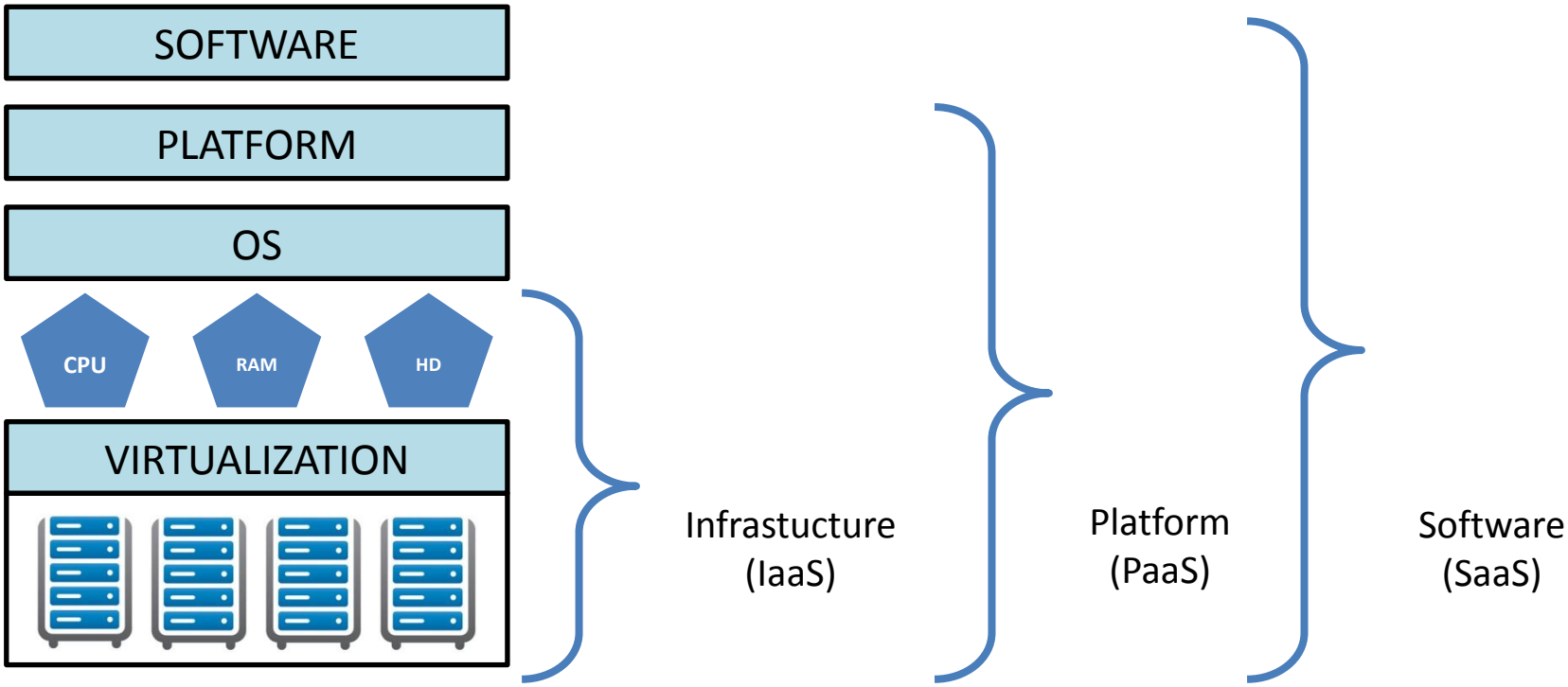
CS 436
Cloud Computing

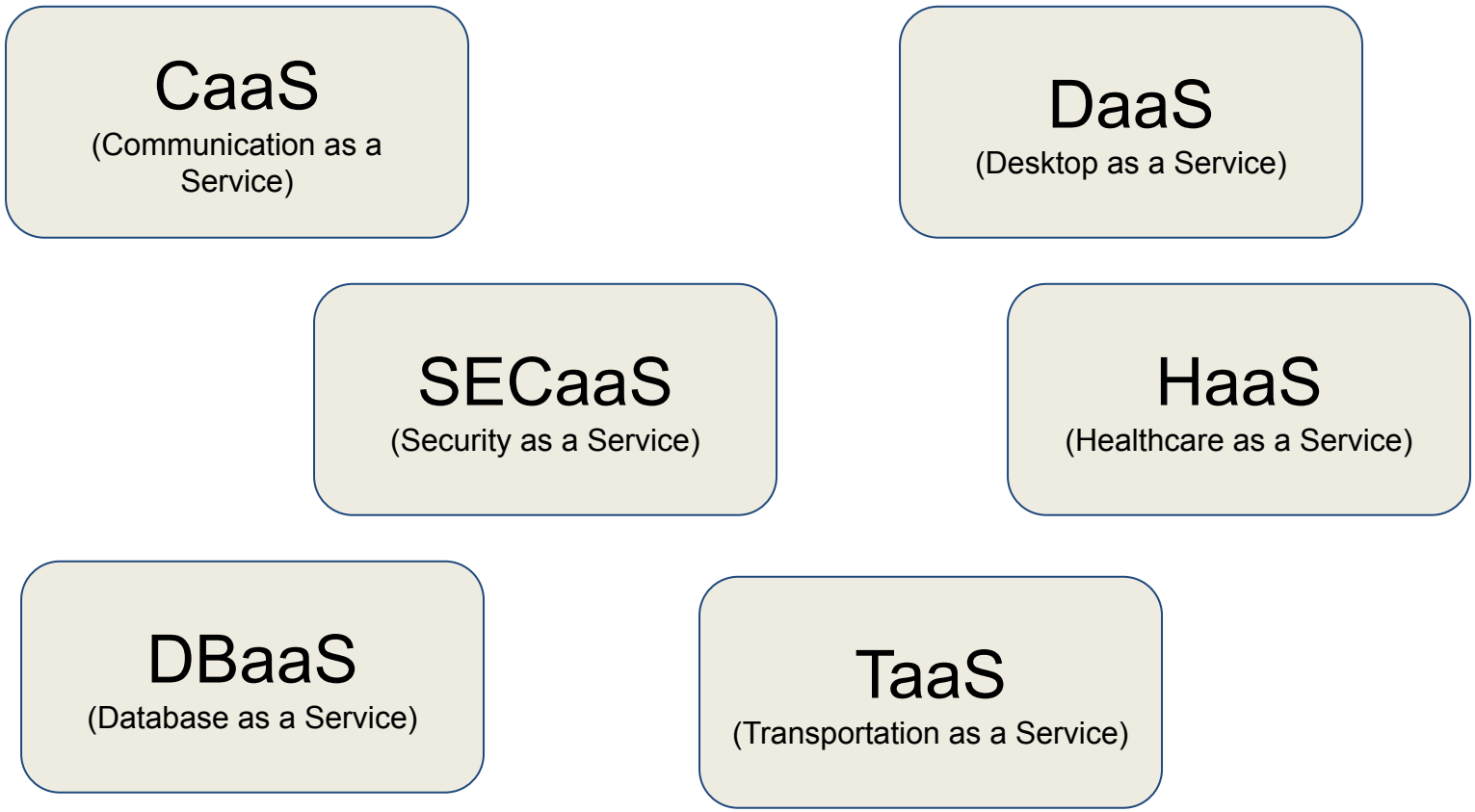
13.02.2025

Cloud Computing Service Models

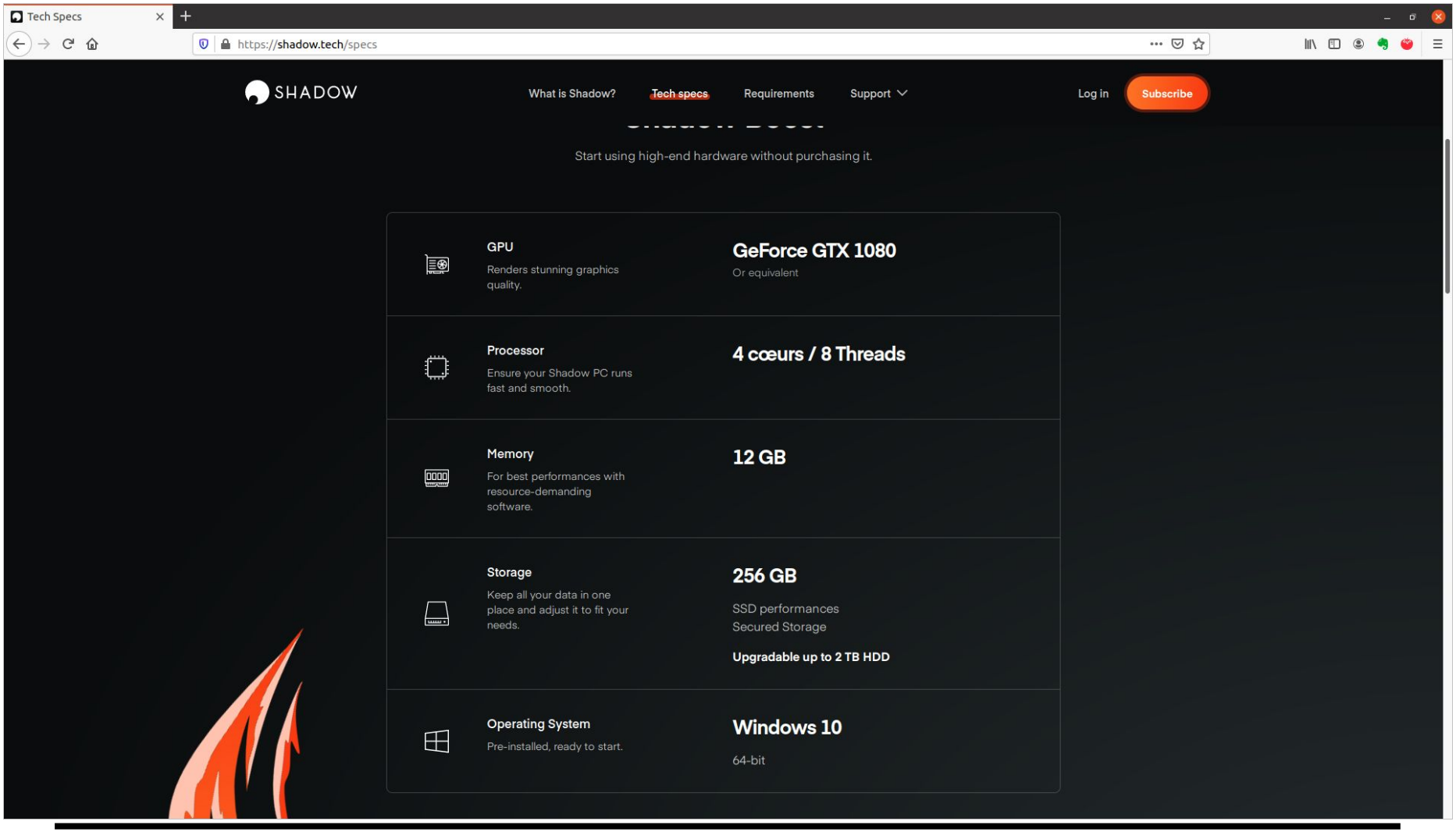
In what forms can I buy this utility (computing) ?



XaaS Anything as a Service



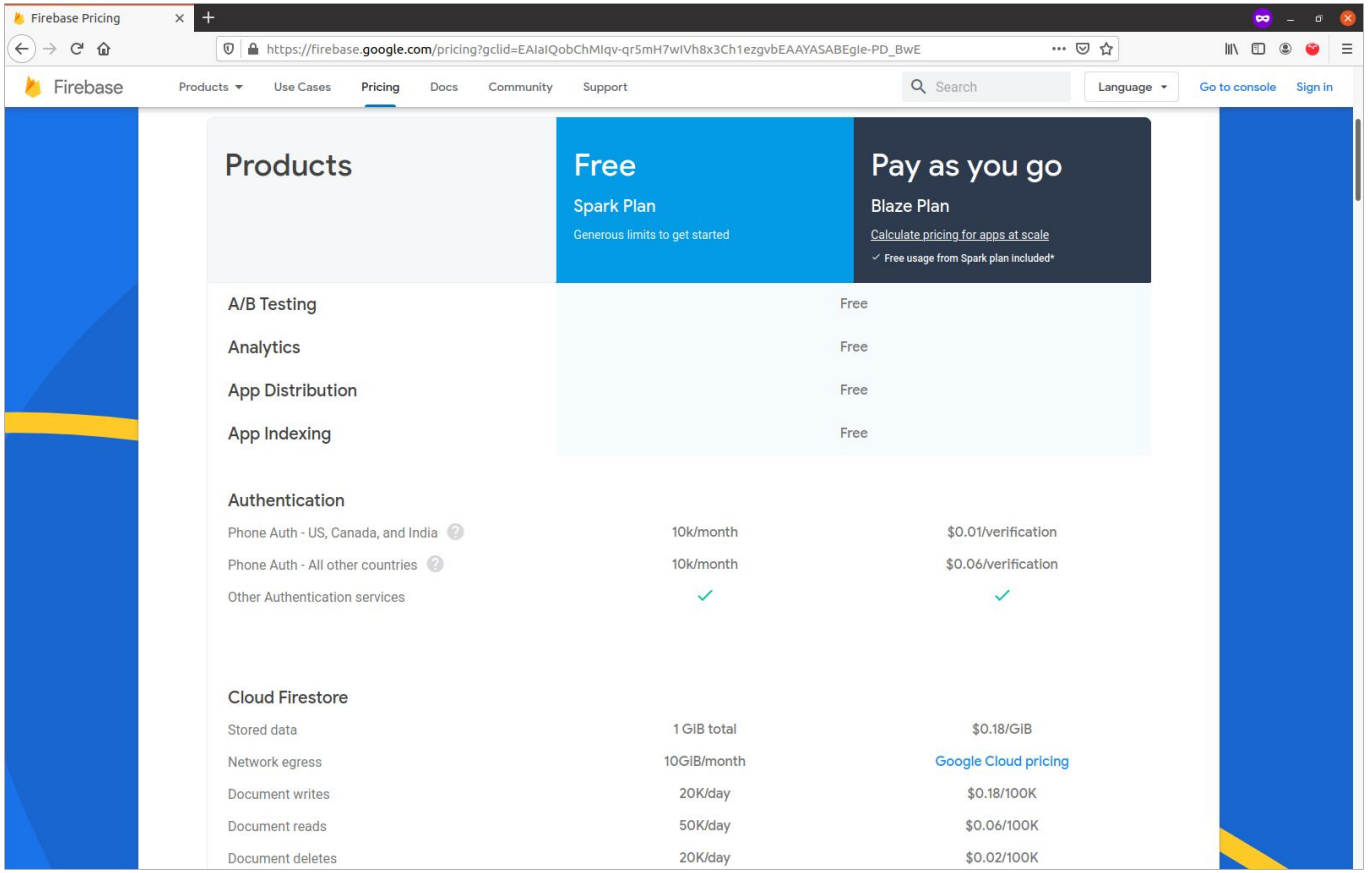
Game Desktop as a Service



Backend as a Service



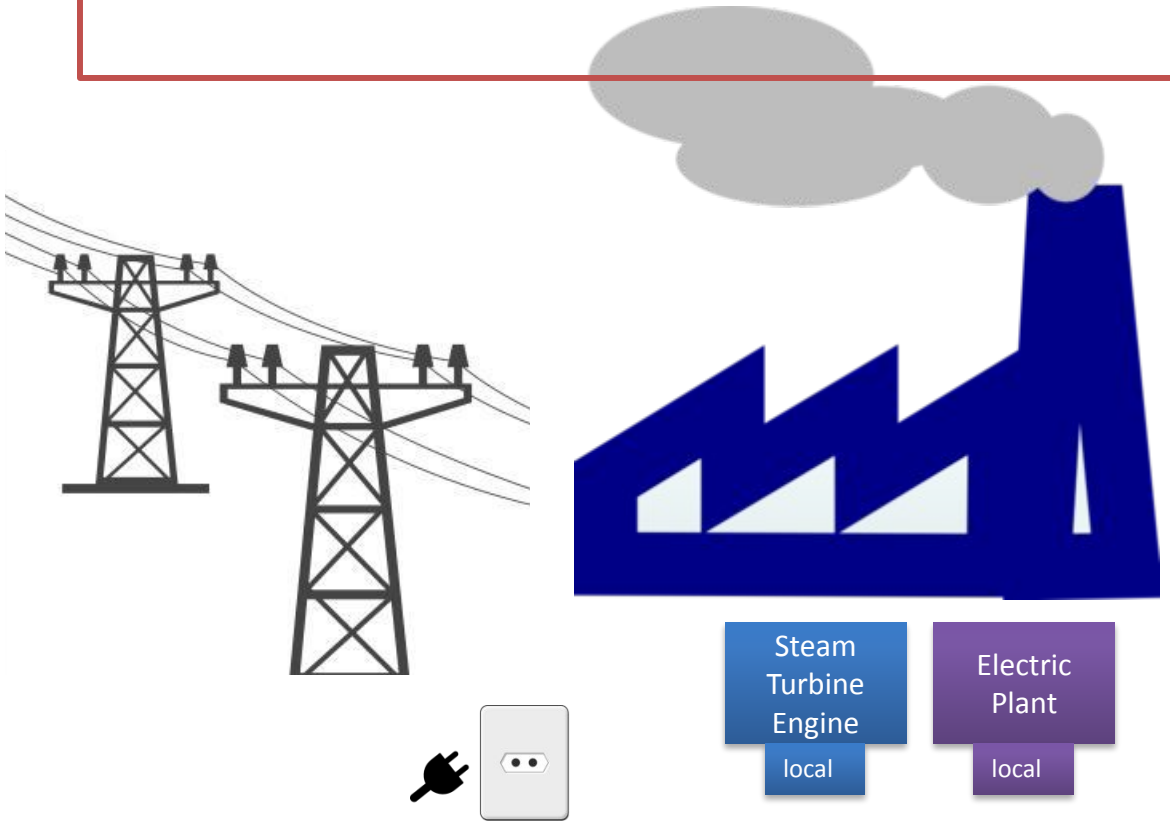
Firebase Pricing



https://firebase.google.com/pricing

Understanding Cloud Computing

Utility Computing (The Electricity Analogy)



Opportunities

- **Scalability** – Cloud platforms allow businesses to scale resources up or down based on demand, enabling cost-effective resource allocation.
- **Flexibility** – Users can choose different configurations (VMs, containers, serverless) and quickly adapt to changing workloads.
- **Global Reach & Physical Redundancy** – Cloud providers have data centers across multiple regions, ensuring availability and failover options.
- **Storage for Compute & Network Resources** – Elastic storage options (block, object, file storage) enable efficient handling of different workloads.
- **Reliability & Redundancy** – Cloud providers implement data replication, backup, and fault tolerance mechanisms to enhance system availability.

Opportunities

- **Security Features** – Cloud platforms provide built-in security tools such as encryption, IAM, and compliance certifications.
- **Automated Software Updates** – Cloud providers handle software patches, reducing administrative burden.
- **Cost Efficiency (Pay-as-you-go)** – The cloud allows organizations to pay only for what they use, reducing upfront capital expenditures.
- **Simplified Maintenance** – Managed services reduce the need for in-house maintenance efforts.

Challenges

Limited Control over Infrastructure – Users rely on cloud providers for hardware, networking, and low-level configurations.
Strict Environmental Conditions (Cooling, Power, A/C) – While cloud providers handle this, edge computing or on-premises cloud setups may require strict environmental management.
Administrative Overhead – Managing IAM policies, monitoring usage, and setting up security controls can be complex.
Component Failures – Hardware failures can still impact cloud resources, requiring high-availability architectures.
Data Reliability & Compliance – Ensuring data consistency, GDPR compliance, and secure backups requires careful planning.

Challenges

Security Concerns – Misconfigurations, unauthorized access, and API vulnerabilities remain risks in cloud environments.
Cost Model Complexity – “Pay-as-you-go” provides flexibility, but unexpected costs (egress fees, reserved instances) can make budgeting challenging.
Complicated Architecture – Multi-cloud and hybrid cloud setups introduce complexity in integration, monitoring, and orchestration.
Vendor Lock-in – Moving workloads between providers can be difficult due to proprietary APIs, data migration costs, and dependency on specific services.